

**Time:** 5 minutes**Outcome:** Apply open-market operations, administered rates, and reserve tools.**Difficulty:**

Intermediate

Monetary-Policy Tools



THE CORE IDEA

The Federal Reserve influences bank reserves, the money supply, and short-term interest rates through several policy tools. Open-market purchases add reserves to the banking system, while open-market sales remove them. Administered rates change the incentives banks face when borrowing or holding reserves. Changes in reserves can affect bank lending and deposit creation, although the actual expansion of money may be smaller than the simple textbook multiplier predicts.



HOW TO RECOGNIZE IT

- An open-market purchase adds reserves; in the limited-reserves model this tends to lower short-term rates.
- An open-market sale removes reserves; in the limited-reserves model this tends to raise short-term rates.
- Lower administered rates ease monetary conditions; higher administered rates restrain them.
- In an ample-reserves system, administered rates are the main tool for controlling short-term rates.



WATCH OUT

More reserves do not guarantee a fixed expansion of broad money. Banks, borrowers, and depositors affect the response; the simple multiplier is a separate restrictive model.



WORKED EXAMPLE: MATCHING THE TOOL TO ITS EFFECT

Policy action and immediate textbook effect

	Immediate effect
Open-market purchase	Adds bank reserves; can lower rates when reserves are limited.
Open-market sale	Removes reserves; can raise rates when reserves are limited.
Lower administered rates	Lowers the rates guiding short-term market rates.
Higher administered rates	Raises the rates guiding short-term market rates.

A \$50 million security purchase adds \$50 million of reserves. In a limited-reserves textbook market, that tends to lower short-term interest rates. In an ample-reserves system, administered rates are the main rate-control tool. Distinguish each tool's immediate effect from the eventual lending and spending response.



CHECK YOURSELF

If the central bank lowers the rate it charges banks for direct loans, what is the immediate textbook effect?

**READY?**

Return to the game and master this concept.